

_osmnx-enrich

November 17, 2025

```
[1]: %reload_ext autoreload  
     %autoreload 2
```

```
[2]: from waymo_agent import *
```

```
[26]: from waymo_agent.osmnx.inspect import *  
      from waymo_agent.osmnx.osmnx_constants import *  
      from waymo_agent.osmnx.sparsify import *  
      from waymo_agent.osmnx.utils import _max_speed_int  
      from waymo_agent.osmnx.post_process_enrichment import *
```

Load Sparse Graph

```
[4]: config = EnvConfig()
```

```
[5]: G = ox.load_graphml(filepath=MAP_DIR / config.map_name)
```

```
[8]: edge_colors, edge_widths = OSMNXConstants.get_edge_colors(G)  
     fig, ax = ox.plot_graph(  
         G,  
         edge_color=edge_colors,  
         edge_linewidth=edge_widths,  
         **{**OX_PLOT_DEFAULTS | {"figsize": (6, 6)}},  
     )
```



```
[14]: OX_PLOT_DEFAULTS
```

```
[14]: {'node_size': 5,  
      'node_color': 'white',  
      'node_alpha': 0.5,  
      'figsize': (10, 10),  
      'bgcolor': 'black',  
      'show': True}
```

0.1 Enrich Graph

```
[49]: enrich_graph(G)
```

Assigned 18 charging nodes.

```
[56]: inspect_graph(G)
```

Graph has 772 nodes and 1609 edges.

Sample node attributes: {'y': 40.7756143, 'x': -73.9604288, 'highway':

```
'traffic_signals', 'street_count': 4, 'is_charger': False, 'lambda': 0.1578,
'num_chargers': False}
```

```
Sample edge attributes: {'osmid': 458177128, 'highway': 'secondary', 'lanes':
'3', 'maxspeed': '25', 'name': 'East 37th Street', 'oneway': True, 'reversed':
False, 'length': 147.1120974719425, 'geometry': <LINESTRING (-73.978 40.748,
-73.978 40.748, -73.98 40.749, -73.98 40.749, -...>}
```

```
[51]: sum_lambda = sum_graph_attr(G, "lambda", "node")
      round(sum_lambda, 2)
```

```
[51]: 77.2
```

```
[52]: sum_chargers = sum_graph_attr(G, "num_chargers", "node")
      round(sum_chargers, 2)
```

```
[52]: 18.0
```

```
[ ]: ## Plot Enriched Graph
      ## TODO color node by whether it has chargers or not
      ## TODO edge color by speed limit
      ## TODO node size by lambda
      edge_colors, edge_widths = OSMNXConstants.get_edge_colors(G)
      fig, ax = ox.plot_graph(
          G,
          edge_color=edge_colors,
          edge_linewidth=edge_widths,
          **{**OX_PLOT_DEFAULTS | {"figsize": (6, 6)}},
      )
```